

Final Exam Solutions

Riemannian and Complex Geometry

Arsh Dhawan

24 January 2026

Physics LATAM

Lecturers: Daniel Galviz & Bowen Liu

Teaching Assistants: Anis Bousclet & Julian Alzate

Problem 1: Spherical Average Formula for Scalar Curvature

Statement. For a Riemannian manifold (M, g) and $p \in M$, show

$$\text{scal}(p) = \frac{n}{\omega_{n-1}} \int_{S^{n-1} \subset T_p M} \text{Ric}_p(x, x) dS,$$

where ω_{n-1} is the area of the unit sphere $S^{n-1} \subset T_p M$ and dS is the area element.

Solution.

Step 1: Setup and component representation

Fix an orthonormal basis $(e_1, \dots, e_n)$ of the tangent space $T_p M$. Define the Ricci tensor components $\text{Ric}_{ij} := \text{Ric}(e_i, e_j)$ and note that the scalar curvature is $\text{scal}(p) = \sum_{i=1}^n \text{Ric}_{ii}$ (the trace of the Ricci tensor).

For any vector $x \in S^{n-1}$ (the unit sphere in $T_p M$), write $x = \sum_{i=1}^n x_i e_i$ with the constraint $\sum_{i=1}^n x_i^2 = 1$. By the definition of the Ricci tensor as a bilinear form,

$$\text{Ric}(x, x) = \left\langle \sum_i x_i e_i, \sum_j x_j e_j \right\rangle_{\text{Ric}} = \sum_{i,j=1}^n \text{Ric}_{ij} x_i x_j.$$

Step 2: Integral decomposition

Substituting the quadratic form into the integral,

$$\int_{S^{n-1}} \text{Ric}(x, x) dS = \int_{S^{n-1}} \sum_{i,j=1}^n \text{Ric}_{ij} x_i x_j dS = \sum_{i,j=1}^n \text{Ric}_{ij} \int_{S^{n-1}} x_i x_j dS.$$

Thus, to evaluate the left-hand side, it suffices to compute the *moment integrals* $\int_{S^{n-1}} x_i x_j dS$ for all i, j .

Step 3: Spherical moment tensor from rotational symmetry

The unit sphere S^{n-1} is invariant under the orthogonal group $O(n)$. This rotational symmetry has a powerful consequence: the tensor $M_{ij} := \int_{S^{n-1}} x_i x_j dS$ must be proportional to the Euclidean metric δ_{ij} . That is,

$$\int_{S^{n-1}} x_i x_j dS = c \cdot \delta_{ij}$$

for some constant c independent of i and j . This holds because any other rank-2 tensor would break rotational symmetry.

To find c , take the trace over all indices:

$$\sum_{i=1}^n \int_{S^{n-1}} x_i x_i dS = \int_{S^{n-1}} \sum_{i=1}^n x_i^2 dS = \int_{S^{n-1}} 1 dS = \omega_{n-1}.$$

On the other hand, using $\int_{S^{n-1}} x_i x_j dS = c \delta_{ij}$, the left-hand side is

$$\sum_{i=1}^n \int_{S^{n-1}} x_i x_i dS = \sum_{i=1}^n c \cdot \delta_{ii} = c \cdot n.$$

Equating $cn = \omega_{n-1}$ yields $c = \omega_{n-1}/n$. Therefore,

$$\int_{S^{n-1}} x_i x_j dS = \frac{\omega_{n-1}}{n} \delta_{ij}.$$

Step 4: Evaluation of the Ricci integral

Substituting the moment formula into the decomposition from Step 2,

$$\int_{S^{n-1}} \text{Ric}(x, x) dS = \sum_{i,j=1}^n \text{Ric}_{ij} \cdot \frac{\omega_{n-1}}{n} \delta_{ij} = \frac{\omega_{n-1}}{n} \sum_{i=1}^n \text{Ric}_{ii} = \frac{\omega_{n-1}}{n} \text{scal}(p).$$

Dividing both sides by ω_{n-1} ,

$$\frac{1}{\omega_{n-1}} \int_{S^{n-1}} \text{Ric}(x, x) dS = \frac{\text{scal}(p)}{n}.$$

Step 5: Conclusion and normalization

Multiplying both sides by n yields:

$$\text{scal}(p) = \frac{n}{\omega_{n-1}} \int_{S^{n-1}} \text{Ric}(x, x) dS.$$

Equivalently, the *average value* (dividing the integral by ω_{n-1}) of $\text{Ric}(x, x)$ over the unit sphere is $\text{scal}(p)/n$. This is the stated formula with the geometric interpretation: **scalar curvature is n times the mean of the Ricci quadratic form over all unit directions**, or equivalently, the average trace of Ricci curvature in all directions.

Remark: Connection to Gaussian measure

The moment result $\int_{S^{n-1}} x_i x_j dS = (\omega_{n-1}/n) \delta_{ij}$ can also be understood as follows. If $X \sim \mathcal{N}(0, I_n)$ is a standard Gaussian in $\mathbb{R}^n$, then $\mathbb{E}[X_i X_j] = \delta_{ij}$ and the unit vector $X/\|X\|$ is uniformly distributed on S^{n-1} . Our formula follows from a simple rescaling argument comparing expectations with respect to the Gaussian measure and the uniform measure on S^{n-1} . $\square$

Problem 3: Lichnerowicz Estimate and Equality Case

Statement. On a compact Riemannian manifold (M, g) of dimension n with $\text{Ric}(g) \geq \lambda g > 0$, show for the first nonzero eigenvalue λ_1 of the Laplacian Δ that

$$\lambda_1 \geq \frac{n}{n-1} \lambda,$$

and equality holds iff $\text{Hess } f = -\frac{\Delta f}{n} g$ for an eigenfunction $\Delta f = -\lambda_1 f$.

Solution.

Convention: We use the *nonpositive Laplacian* convention where $\Delta f = -\text{div}(\nabla f)$, so eigenvalues of Δ are nonpositive and we write $\Delta f = -\lambda_1 f$ with $\lambda_1 > 0$.

Step 1: Bochner formula and integration by parts

Let f be a smooth eigenfunction satisfying $\Delta f = -\lambda_1 f$ with $\int_M f dV_g = 0$ (we can assume f is not identically zero, so $\lambda_1 > 0$). The *Bochner formula* relates the Hessian, the Laplacian, and the Ricci curvature:

$$\frac{1}{2} \Delta |\nabla f|^2 = |\text{Hess } f|^2 + \langle \nabla f, \nabla(\Delta f) \rangle + \text{Ric}(\nabla f, \nabla f).$$

This formula holds for any smooth function on a Riemannian manifold.

Integrate both sides over M with respect to the volume measure dV_g :

$$\frac{1}{2} \int_M \Delta |\nabla f|^2 dV_g = \int_M |\text{Hess } f|^2 dV_g + \int_M \langle \nabla f, \nabla(\Delta f) \rangle dV_g + \int_M \text{Ric}(\nabla f, \nabla f) dV_g.$$

Since M is compact without boundary, integration by parts (or the divergence theorem) applied to the Laplacian gives

$$\int_M \Delta h dV_g = 0$$

for any smooth function h on M . Thus $\int_M \Delta |\nabla f|^2 dV_g = 0$, and the equation becomes

$$\int_M |\text{Hess } f|^2 dV_g + \int_M \langle \nabla f, \nabla(\Delta f) \rangle dV_g + \int_M \text{Ric}(\nabla f, \nabla f) dV_g = 0.$$

Step 2: Substitute the eigenvalue equation

Since $\Delta f = -\lambda_1 f$, we have $\nabla(\Delta f) = -\lambda_1 \nabla f$. Therefore,

$$\int_M \langle \nabla f, \nabla(\Delta f) \rangle dV_g = \int_M \langle \nabla f, -\lambda_1 \nabla f \rangle dV_g = -\lambda_1 \int_M |\nabla f|^2 dV_g.$$

Substituting back,

$$\int_M |\text{Hess } f|^2 dV_g - \lambda_1 \int_M |\nabla f|^2 dV_g + \int_M \text{Ric}(\nabla f, \nabla f) dV_g = 0.$$

Rearranging,

$$\int_M |\text{Hess } f|^2 dV_g + \int_M \text{Ric}(\nabla f, \nabla f) dV_g = \lambda_1 \int_M |\nabla f|^2 dV_g. \quad (*)$$

Step 3: Lower bound on Ricci term

We are given that $\text{Ric}(g) \geq \lambda g$ pointwise, meaning $\text{Ric}(v, v) \geq \lambda g(v, v)$ for all tangent vectors v . Applying this to $v = \nabla f$ (which is a vector field),

$$\text{Ric}(\nabla f, \nabla f) \geq \lambda |\nabla f|^2.$$

Integrating,

$$\int_M \text{Ric}(\nabla f, \nabla f) dV_g \geq \lambda \int_M |\nabla f|^2 dV_g. \quad (**)$$

Step 4: Hessian decomposition and lower bound

Decompose the Hessian of f into its trace and trace-free parts:

$$\text{Hess } f = \underbrace{\left(\text{Hess } f - \frac{1}{n}(\Delta f)g \right)}_{=:\text{Hess}^\circ f \text{ (trace-free)}} + \underbrace{\frac{1}{n}(\Delta f)g}_{\text{trace part}}.$$

Since $\text{Hess}^\circ f$ is trace-free and symmetric, it is orthogonal (with respect to the standard inner product on symmetric $(0, 2)$ tensors) to the trace part. Therefore,

$$|\text{Hess } f|^2 = |\text{Hess}^\circ f|^2 + \left| \frac{1}{n}(\Delta f)g \right|^2.$$

The second term is $|\frac{1}{n}(\Delta f)g|^2 = \frac{1}{n^2}(\Delta f)^2 \cdot |g|^2 = \frac{1}{n^2}(\Delta f)^2 \cdot n = \frac{1}{n}(\Delta f)^2$ (since $|g|^2 = n$ for a metric). Thus,

$$|\text{Hess } f|^2 \geq \frac{1}{n}(\Delta f)^2. \quad (***)$$

Integrating,

$$\int_M |\text{Hess } f|^2 dV_g \geq \frac{1}{n} \int_M (\Delta f)^2 dV_g.$$

Step 5: Connect Laplacian and gradient

For an eigenfunction with $\Delta f = -\lambda_1 f$, use the standard Rayleigh quotient identity. Integration by parts shows

$$\int_M |\nabla f|^2 dV_g = \int_M f \cdot (-\Delta f) dV_g = \int_M f \cdot \lambda_1 f dV_g = \lambda_1 \int_M f^2 dV_g.$$

Similarly,

$$\int_M (\Delta f)^2 dV_g = \int_M (\lambda_1 f)^2 dV_g = \lambda_1^2 \int_M f^2 dV_g.$$

Step 6: Combine inequalities and derive the lower bound

From $(***)$,

$$\int_M |\text{Hess } f|^2 dV_g \geq \frac{1}{n} \int_M (\Delta f)^2 dV_g = \frac{\lambda_1^2}{n} \int_M f^2 dV_g.$$

Substituting into $(*)$ together with $(**)$,

$$\frac{\lambda_1^2}{n} \int_M f^2 dV_g + \lambda \int_M |\nabla f|^2 dV_g \leq \lambda_1 \int_M |\nabla f|^2 dV_g.$$

Using $\int_M |\nabla f|^2 dV_g = \lambda_1 \int_M f^2 dV_g$ from Step 5,

$$\frac{\lambda_1^2}{n} \int_M f^2 dV_g + \lambda \cdot \lambda_1 \int_M f^2 dV_g \leq \lambda_1 \cdot \lambda_1 \int_M f^2 dV_g.$$

Dividing by $\int_M f^2 dV_g > 0$,

$$\frac{\lambda_1^2}{n} + \lambda \lambda_1 \leq \lambda_1^2.$$

Rearranging,

$$\lambda \lambda_1 \leq \lambda_1^2 - \frac{\lambda_1^2}{n} = \lambda_1^2 \left(1 - \frac{1}{n}\right) = \lambda_1^2 \cdot \frac{n-1}{n}.$$

Dividing by $\lambda_1 > 0$,

$$\lambda \leq \lambda_1 \cdot \frac{n-1}{n}.$$

Therefore,

$$\boxed{\lambda_1 \geq \frac{n}{n-1} \lambda}.$$

Step 7: Characterization of equality

Equality $\lambda_1 = \frac{n}{n-1} \lambda$ holds if and only if all inequalities in the derivation are tight. This requires:

1. Equality in $(**)$: $\text{Ric}(\nabla f, \nabla f) = \lambda |\nabla f|^2$ everywhere.
2. Equality in $(***)$: $\text{Hess}^\circ f \equiv 0$ (i.e., the trace-free part of the Hessian vanishes).

Condition (2) means $\text{Hess } f$ is purely trace:

$$\text{Hess } f = \frac{1}{n}(\Delta f)g = \frac{1}{n}(-\lambda_1 f)g = -\frac{\lambda_1}{n} f \cdot g.$$

This is equivalent to $\text{Hess } f = -\frac{\Delta f}{n} g$, which is the stated condition. In coordinate-free terms, this means the Hessian of f is conformal to the metric g (proportional to g at each point), which is a highly restrictive geometric condition that characterizes extremal eigenfunctions. $\square$

Problem 4: Injectivity of $f : H_{\text{dR}}^1(X, \mathbb{R}) \rightarrow H^{0,1}(X)$

Statement. On a Hermitian manifold (X, ω) , the map $f([\eta]) = [\eta^{0,1}]$ is an injective $\mathbb{R}$ -linear map.

Solution.

Step 1: Type decomposition of differential forms

On a complex manifold X of complex dimension m (hence real dimension $n = 2m$), any differential k -form can be uniquely written as a sum of forms of type (p, q) with $p + q = k$. In particular, any real 1-form η can be decomposed as

$$\eta = \eta^{1,0} + \eta^{0,1},$$

where $\eta^{1,0}$ is a $(1,0)$ -form (picks out the holomorphic part) and $\eta^{0,1}$ is a $(0,1)$ -form (picks out the antiholomorphic part).

Because η is real (i.e., $\bar{\eta} = \eta$), the two components are **complex conjugates** of each other:

$$\eta^{1,0} = \overline{\eta^{0,1}} \quad \text{and} \quad \eta^{0,1} = \overline{\eta^{1,0}}.$$

To derive this, decompose both η and its conjugate:

$$\eta = \eta^{1,0} + \eta^{0,1}, \quad \bar{\eta} = \overline{\eta^{1,0}} + \overline{\eta^{0,1}}.$$

Since η is real, $\bar{\eta} = \eta$, so

$$\eta^{1,0} + \eta^{0,1} = \overline{\eta^{1,0}} + \overline{\eta^{0,1}}.$$

By uniqueness of the (p, q) decomposition of forms, we conclude $\eta^{1,0} = \overline{\eta^{0,1}}$ and $\eta^{0,1} = \overline{\eta^{1,0}}$.

Step 2: The map f is well-defined

We claim that f sends de Rham classes to Dolbeault classes. Let $[\eta] \in H_{\text{dR}}^1(X, \mathbb{R})$ be a real closed 1-form (so $d\eta = 0$). If we replace η by $\eta' = \eta + d\varphi$ where φ is a real function, then

$$(\eta')^{0,1} = (\eta + d\varphi)^{0,1} = \eta^{0,1} + (d\varphi)^{0,1}.$$

Since φ is real and of degree 0, we have

$$(d\varphi)^{0,1} = \bar{\partial}\varphi.$$

This is because $d\varphi = \partial\varphi + \bar{\partial}\varphi$ (the Dolbeault decomposition of the exterior derivative), and $\partial\varphi$ is $(1,0)$ while $\bar{\partial}\varphi$ is $(0,1)$. Therefore,

$$[\eta^{0,1}]_{\bar{\partial}} = [(\eta')^{0,1}]_{\bar{\partial}}.$$

Thus, $f([\eta]) = [\eta^{0,1}]$ depends only on the de Rham class of η , not on the choice of representative.

Step 3: The map f is $\mathbb{R}$ -linear

For real scalars $a, b \in \mathbb{R}$ and real closed 1-forms η_1, η_2 ,

$$f([a\eta_1 + b\eta_2]) = [(a\eta_1 + b\eta_2)^{0,1}] = [a\eta_1^{0,1} + b\eta_2^{0,1}] = a[\eta_1^{0,1}] + b[\eta_2^{0,1}] = af([\eta_1]) + bf([\eta_2]).$$

This follows from the linearity of type decomposition and the linearity of Dolbeault cohomology.

Step 4: Injectivity via kernel analysis

Suppose $[\eta] \in H_{\text{dR}}^1(X, \mathbb{R})$ satisfies $f([\eta]) = 0$, meaning $[\eta^{0,1}] = 0$ in $H^{0,1}(X)$ (the first Dolbeault cohomology group). By definition of this cohomology group, if $\eta^{0,1}$ represents the zero class, then there exists a (complex-valued) smooth function ψ on X such that

$$\eta^{0,1} = \bar{\partial}\psi.$$

Now use the fact that η is real. From Step 1, the two components are complex conjugates: $\eta^{1,0} = \overline{\eta^{0,1}}$. Since $\eta^{0,1} = \bar{\partial}\psi$, we have

$$\eta^{1,0} = \overline{\eta^{0,1}} = \overline{\bar{\partial}\psi}.$$

By the conjugation property of Dolbeault operators on complex manifolds, complex conjugation swaps the operators and conjugates the argument:

$$\overline{\bar{\partial}\psi} = \partial\bar{\psi}.$$

Therefore,

$$\eta^{1,0} = \partial\bar{\psi}.$$

Step 5: Recovery of η as an exact real form

Combining the results from Step 4,

$$\eta = \eta^{1,0} + \eta^{0,1} = \partial\bar{\psi} + \bar{\partial}\psi.$$

To show this is exact, decompose ψ into its real and imaginary parts. Write $\psi = u + iv$ where u and v are real smooth functions. Then $\bar{\psi} = u - iv$ (complex conjugate). Compute each term:

$$\partial\bar{\psi} = \partial(u - iv) = \partial u - i\partial v,$$

$$\bar{\partial}\psi = \bar{\partial}(u + iv) = \bar{\partial}u + i\bar{\partial}v.$$

Adding these,

$$\eta = \partial u - i\partial v + \bar{\partial}u + i\bar{\partial}v = (\partial + \bar{\partial})u + i(\bar{\partial}v - \partial v) = du + i(\bar{\partial}v - \partial v).$$

Since η is real and $d = \partial + \bar{\partial}$, the imaginary part must vanish: $\bar{\partial}v - \partial v = 0$, or $\bar{\partial}v = \partial v$. Moreover, for any real function v , we have $dv = \partial v + \bar{\partial}v$ is real, which means ∂v and $\bar{\partial}v$ are complex conjugates. The only way their difference equals zero is if both vanish: $\partial v = 0$ and $\bar{\partial}v = 0$. Therefore,

$$\eta = du,$$

which shows η is exact in de Rham cohomology.

Step 6: Conclude kernel is trivial

From Step 5, if $[\eta^{0,1}] = 0$, then $\eta = du$ for some real function u . Therefore, $[\eta] = 0$ in $H_{\text{dR}}^1(X, \mathbb{R})$. This proves that the kernel of f is trivial, so f is injective.

Step 7: Consequence for cohomology dimensions

Injectivity of f implies

$$\dim_{\mathbb{R}} H_{\text{dR}}^1(X, \mathbb{R}) \leq \dim_{\mathbb{C}} H^{0,1}(X).$$

This is a non-obvious relation between de Rham and Dolbeault cohomologies on Hermitian (not necessarily Kähler) manifolds. For a Kähler manifold, stronger Hodge decomposition theorems hold, but this result applies more generally. $\square$